POWER SOLUTIONS

Technical Engineering Manual

Off-Grid Solar Power Systems for Remote Equipment

WP-384 | WP-768 | WP-4800T | WP-21600T

Document: WC-PS-TEM-v1.2 | Revision: February 2026

FOR ENGINEERING USE

Revision History

Version	Date	Author	Description
1.0	February 2026	Hardware & Firmware Department	Initial release

Table of Contents

1. Introduction and Scope

1.1 Document Purpose

This Technical Engineering Manual provides comprehensive engineering documentation for the WakeCap Power Solutions product line. It is intended for engineers requiring detailed technical understanding of system design, performance characteristics, and integration requirements.

1.2 Product Family Overview

The WakeCap Power Solutions are autonomous off-grid solar power systems designed to provide reliable DC power for remote equipment in harsh environments. The product family comprises four tiers based on energy capacity:

Product	Model	Battery	Solar	System Voltage	Target Load
WakeCap Power 384	WP-384	30Ah $LiFePO_4$	80W	12V DC	≤8W
WakeCap Power 768	WP-768	60Ah $LiFePO_4$	100W	12V DC	≤15W
WakeCap Power 4800T	WP-4800T	400Ah GEL	450W	12V DC	≤100W
WakeCap Power 21600T	WP-21600T	900Ah Lead Acid	1305W	24V DC	≤350W

1.3 Design Environment

All systems are designed for deployment in Saudi Arabia with the following environmental considerations:

- High ambient temperatures (up to 55°C)
- Desert conditions with high dust/sand exposure
- Coastal areas with salt-laden humid air
- High solar irradiance (Peak Sun Hours: 5.5-7.5 hours)
- Remote locations with limited grid access

1.4 Applicable Standards

- IEC 62124: Photovoltaic (PV) standalone systems - Design verification
- IEC 61427: Secondary cells for solar photovoltaic energy systems
- IEC 62509: Battery charge controllers for photovoltaic systems
- IP ratings per IEC 60529
- [TBD - Additional certifications]

2. System Architecture

2.1 Functional Block Diagram

All WakeCap Power Solutions share a common functional architecture:

[IMAGE: System block diagram showing: Solar Panel Array → MPPT Controller → Battery Bank → Load Output with protection circuits]

Key functional blocks:

- Solar Photovoltaic Array: Energy harvesting from solar radiation
- MPPT Charge Controller: Maximum Power Point Tracking and charge regulation
- Battery Energy Storage: Electrochemical energy storage for autonomy
- DC Output Stage: Voltage regulation and load distribution
- Protection Systems: Over-current, over-voltage, temperature, and reverse polarity
- Monitoring/Indicators: State of charge, charging status, fault indication

2.2 Energy Flow

The system operates in three primary modes:

2.2.1 Solar Charging Mode

When solar irradiance is sufficient (typically >200 W/m^2):

- MPPT controller tracks maximum power point of solar array
- Solar energy charges battery and simultaneously powers load
- Excess energy stored in battery bank
- Charging follows CC-CV profile for battery chemistry

2.2.2 Battery Discharge Mode

When solar input is insufficient (night/overcast):

- Battery supplies full load demand
- Depth of Discharge (DoD) limited to protect battery life
- Low voltage disconnect prevents deep discharge

2.2.3 Float/Maintenance Mode

When battery is fully charged:

- MPPT controller maintains float voltage
- Solar directly powers load with minimal battery cycling
- Periodic equalization charge (lead-acid systems only)

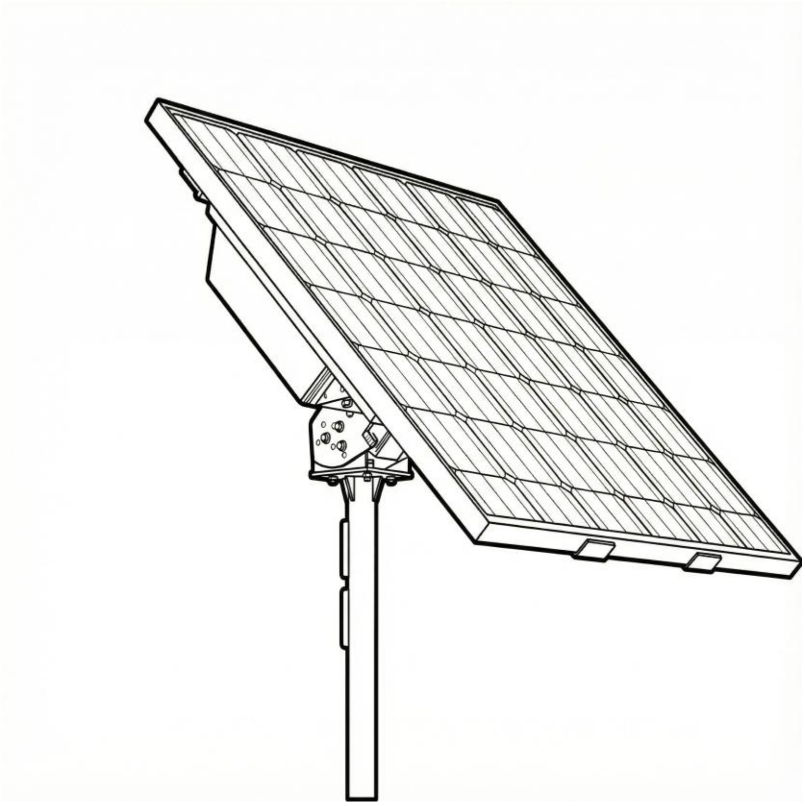
2.3 System Sizing Philosophy

All WakeCap Power Solutions are sized according to:

Design Parameter	Value	Rationale
Backup Duration	48 hours	Two consecutive days without solar (worst case)
Depth of Discharge	70%	Balance between capacity utilization and cycle life
Solar Derating	75%	Account for dust, misalignment, temperature, aging
System Efficiency	90%	Wiring losses, controller self-consumption
DC-DC Efficiency	80%	Conversion losses if voltage conversion required
Peak Sun Hours	6.5h	Annual average for Saudi Arabia

2.4 Architecture by Product

2.4.1 WP-384 Architecture



The WP-384 features a highly integrated design with:

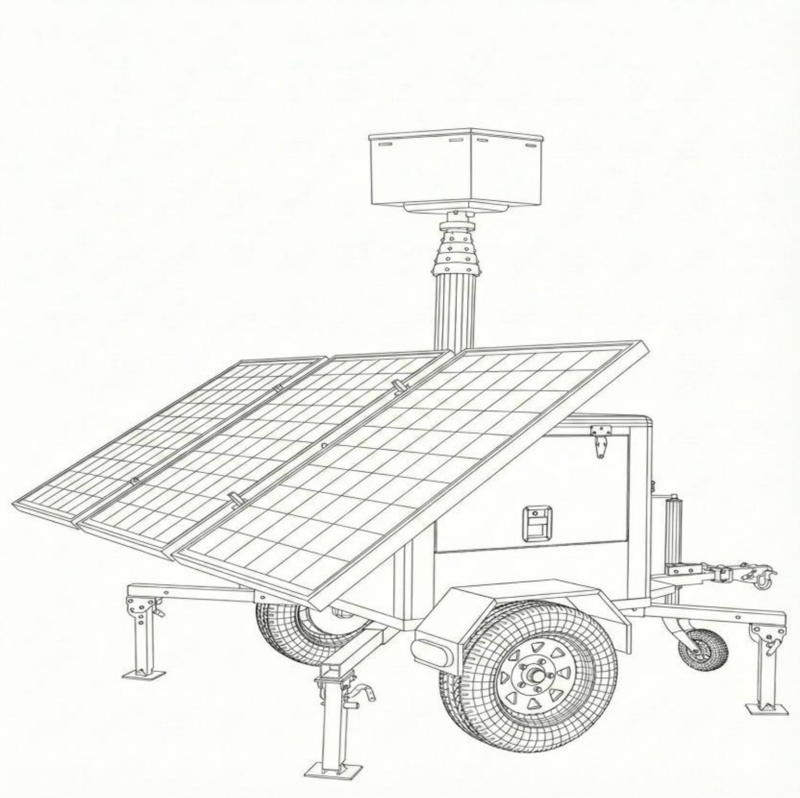
- Aluminum-potted MPPT controller (>95% efficiency) integrated with BMS
- $LiFePO_4$ battery pack with 2P4S cell configuration (30Ah @ 12.8V nominal)
- Integrated SOC indication via LED display
- Single 12V DC output channel with over-current protection
- Optional RS485 interface for remote monitoring

2.4.2 WP-768 Architecture

The WP-768 provides a modular pole-mount design:

- Separate 10A MPPT charge controller
- 60Ah $LiFePO_4$ battery with integrated BMS
- Charging protection board with multiple safety features
- Aviation-style waterproof connectors
- DC output via standard 5.5×2.1mm barrel connector

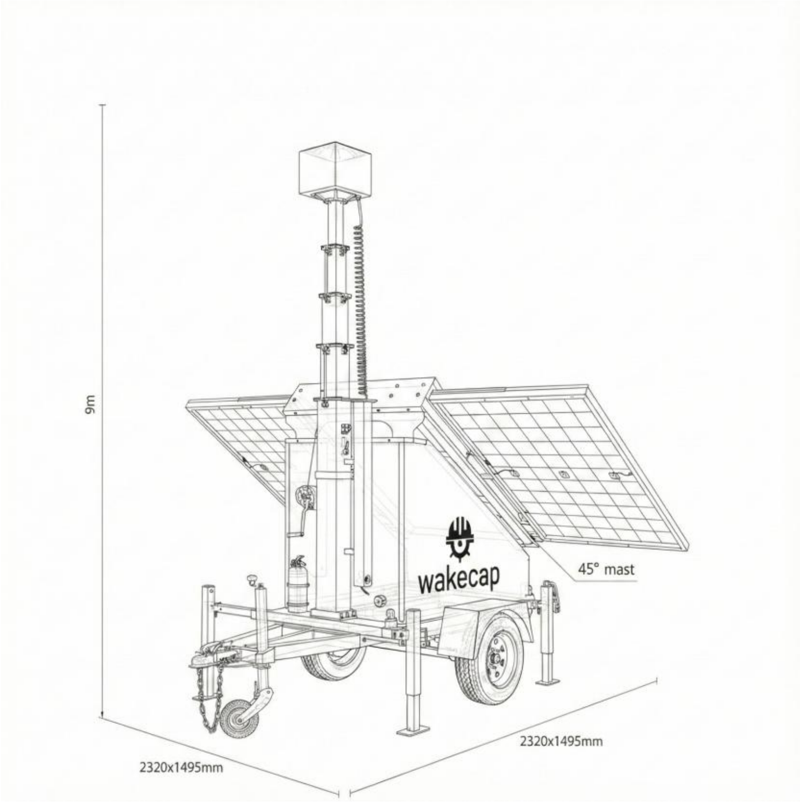
2.4.3 WP-4800T Architecture



The WP-4800T is a trailer-mounted mobile power station:

- 30A MPPT charge controller
- 2×200Ah GEL deep-cycle batteries in parallel (400Ah total)
- 6.5m pneumatic telescoping mast
- Integrated equipment cabinet with thermal management
- Exchange board for data/power distribution
- Electric fan for enclosure ventilation

2.4.4 WP-21600T Architecture



The WP-21600T is the highest-capacity trailer solution:

- EPEVER 60A MPPT charge controller
- 6×150Ah high-temperature lead-acid batteries (3S2P for 24V, 900Ah)
- 9m manual telescoping mast tower
- Large equipment cabinet with multiple cable passages
- CAT6 and power cable routing through mast

3. Design Parameters and Calculations

3.1 Fundamental Equations

System sizing follows the calculation sequence: LOAD → BATTERY → SOLAR → MPPT → AUXILIARY

3.1.1 Battery Capacity Calculation

Required battery capacity to support a given load for specified backup duration:

$$B = (T \times P) / (DoD \times \eta_{sys} \times \eta_{conv} \times V)$$

Where:

- B = Required battery capacity (Ah)
- T = Backup time (hours) = 48
- P = Total system load power (W)
- DoD = Depth of Discharge = 0.70
- η_{sys} = System efficiency = 0.90
- η_{conv} = Converter efficiency = 0.80
- V = System voltage (V)

3.1.2 Solar Array Sizing

Required solar panel wattage to fully recharge the battery in one day:

$$S = (B \times V) / (PSH \times \eta_{solar} \times \eta_{mppt})$$

Where:

- S = Required solar array power (W)
- B = Battery capacity (Ah)
- V = System voltage (V)
- PSH = Peak Sun Hours = 6.5
- η_{solar} = Solar derating factor = 0.75
- η_{mppt} = MPPT efficiency = 0.95

3.1.3 MPPT Controller Sizing

Minimum charge controller current rating:

$$I_{mppt} = S / (\eta_{mppt} \times V)$$

Where:

- I_{mppt} = MPPT controller current rating (A)
- S = Solar array power (W)
- η_{mppt} = MPPT efficiency = 0.95
- V = System voltage (V)

3.2 Design Verification Calculations

3.2.1 WP-384 Design Verification

Target Load: 8W continuous

Battery Calculation:

- B = (48 × 8) / (0.70 × 0.90 × 0.80 × 12)
- B = 384 / 6.048 = 63.5 Ah (theoretical)
- Actual: 30Ah (provides ~22h backup at 8W, or 48h at ~4W)
- Note: $LiFePO_4$ chemistry allows higher effective DoD

Solar Calculation:

- S = (30 × 12) / (6.5 × 0.75 × 0.95)
- S = 360 / 4.63 = 77.8W
- Actual: 80W (sufficient)

3.2.2 WP-768 Design Verification

Target Load: 15W continuous

Battery Calculation:

- B = (48 × 15) / (0.70 × 0.90 × 0.80 × 12)
- B = 720 / 6.048 = 119 Ah (theoretical)
- Actual: 60Ah (provides ~24h backup at 15W)

Solar Calculation:

- S = (60 × 12) / (6.5 × 0.75 × 0.95)
- S = 720 / 4.63 = 155.5W
- Actual: 100W (marginal - relies on $LiFePO_4$ efficiency)

3.2.3 WP-4800T Design Verification

Target Load: 100W continuous

Battery Calculation:

- B = (48 × 100) / (0.70 × 0.90 × 0.80 × 12)
- B = 4800 / 6.048 = 794 Ah (theoretical)
- Actual: 400Ah (provides ~24h backup at 100W)

Solar Calculation:

- S = (400 × 12) / (6.5 × 0.75 × 0.95)
- S = 4800 / 4.63 = 1037W

- Actual: 450W (requires multiple sunny days for full recovery)

3.2.4 WP-21600T Design Verification

Target Load: 350W continuous

Battery Calculation:

- B = (48 × 350) / (0.70 × 0.90 × 0.80 × 24)
- B = 16800 / 12.096 = 1389 Ah (theoretical at 24V)
- Actual: 900Ah @ 24V (provides ~30h backup at 350W)

Solar Calculation:

- S = (900 × 24) / (6.5 × 0.75 × 0.95)
- S = 21600 / 4.63 = 4665W
- Actual: 1305W (sized for typical loads of 150-200W)

3.3 Load Capacity Summary

Based on design calculations, the actual continuous load capacity per solution:

Solution	Battery	Max Load (48h)	Max Load (24h)	Max Load (12h)	Practical Limit
WP-384	30Ah/384Wh	~4W	~8W	~16W	8W
WP-768	60Ah/768Wh	~8W	~15W	~30W	15W
WP-4800T	400Ah/4800Wh	~50W	~100W	~200W	100W
WP-21600T	900Ah@24V/21600Wh	~175W	~350W	~700W	350W

⚠ NOTICE

The "Practical Limit" values assume good solar conditions and should be used as the design target. Exceeding these values will reduce backup time below 24 hours during extended overcast periods.

4. Product Technical Specifications

4.1 WakeCap Power 384 (WP-384) Specifications

Model: G8030 | 80W Solar, 384Wh $LiFePO_4$ Off-Grid Power System

4.1.1 Solar Panel Specifications

Parameter	Value	Unit
Panel Material	Grade-A Monocrystalline Silicon	—
Total Output Power	80	W
Panel Configuration	2 × 40W panels	—
Photoelectric Conversion Efficiency	≥23	%
Rated Output Voltage	18	V
Maximum Power Point Voltage (Vmp)	18	V
Open Circuit Voltage (Voc)	24	V
Maximum Power Point Current (Imp)	4.44	A
Short Circuit Current (Isc)	5	A
Panel Dimensions	390 × 530 × 17 (×2 panels)	mm
Panel Weight	3.2	kg
Frame Material	Anodized Aluminum Alloy, 0.8mm	—
Encapsulation	Tempered Glass / EVA / Cells / EVA Backsheet	—
Snow Load Rating	5400	Pa
Wind Load Rating	2400	Pa

4.1.2 Battery Pack Specifications

Parameter	Value	Unit
Cell Type	$LiFePO_4$ (Lithium Iron Phosphate)	—
Cell Model	IFR32140E15.0Ah	—
Cell Configuration	2P4S (2 Parallel, 4 Series)	—
Nominal Voltage	12.8	V
Pack Capacity	30Ah / 384Wh	—
Discharge Voltage Range	11.2 - 14.8	V
Over-Discharge Protection	≤10 ±0.1	V
Over-Charge Protection	>14.6 ±0.1	V
Maximum Charge Current	7	A
High-Temp Protection (Charge/Discharge)	≥60°C prohibited	—

Low-Temp Charge Protection	-10 ±1	°C
Low-Temp Charge Recovery	0 ±1	°C
Low-Temp Discharge Protection	-20 ±1	°C
Low-Temp Discharge Recovery	-15 ±1	°C
SOC Accuracy	<5	%
BMS Communication	IIC	—
Battery Arm Dimensions	535 × 80 × 100	mm
Battery Pack Weight	~2.25	kg

4.1.3 Main Unit Specifications

Parameter	Value	Unit
Model	TD-G8030	—
DC Output Channels	1 (86cm cable, black)	—
Output Voltage	12 ±0.5	V DC
Output Current (Max)	≤3 ±0.05	A
Output Protection Current	>4A for >3 seconds	—
Input Channels	1 (male plug, 220mm cable)	—
Input Voltage Range	15 - 21	V DC
Maximum Input Current	7	A
RS-485 Channels	1 (2-pin terminal, 30cm)	—
Energy Efficiency (3A load)	Max 92	%
Short-Circuit Protection	>5A instantaneous	—
Operating Temperature	-20 to +50	°C
Storage Temperature	-40 to +60	°C
Operating Humidity	15 - 85	% RH
Protection Rating	IP66	—
Factory State of Charge	40 - 50	%
Runtime (no solar, 150mA)	~8-10 days (25°C)	—
Recharge Time (0-100%)	~8-10 hours	—
Installation Method	Pole Mount / Wall Mount	—

4.1.4 LED Indicator Status

Indicator	Flashing	Steady On	Off
Blue Light	Charging	Charge Complete	No Solar Input
Green Light	Overload/Short Circuit	Output Normal	Shutdown/Fault
Red Light	Low Battery	—	Normal Battery

4.2 WakeCap Power 768 (WP-768) Specifications

Model: TS-100W60AH | 100W Solar, 768Wh $LiFePO_4$ Power System

4.2.1 Solar Panel Specifications

Parameter	Value	Unit
Panel Material	N-Type A+ Grade Monocrystalline Silicon	—
Total Output Power	100	W
Panel Configuration	2 × 50W panels	—
Conversion Efficiency	>22	%
Panel Dimensions	390 × 510 (×2 panels)	mm
Frame Material	Anodized Aluminum	—
Glass	3.2mm Tempered	—
Protection Rating	IP66	—
Operating Temperature	-30 to +85	°C
Maximum Operating Current	7.8	A
Open Circuit Voltage	20	V
Maximum Operating Voltage	17	V
Surface Pressure Resistance	60m/s (200kg/m^2)	—

4.2.2 Battery Pack Specifications

Parameter	Value	Unit
Battery Type	$LiFePO_4$ (Lithium Iron Phosphate)	—
Nominal Capacity	60	Ah
Rated Voltage	10.5 - 12.6	V DC
Charging Temperature	-30 to +85	°C
Discharging Temperature	-30 to +85	°C
Protection Features	High-temp, Low-temp compensation, Balance, OC/OD	—

4.2.3 Controller Specifications

Parameter	Value	Unit
Maximum Charging Current	10	A
Maximum Output Current	10	A
Reverse Connection Protection	Yes	—
Controller Self-consumption	0.06	W
Operating Temperature	-30 to +85	°C
MPPT Efficiency	96.5	%

4.2.4 System Specifications

Parameter	Value	Unit
System Weight	23	kg
Packaging Size	940 × 780 × 240	mm
Packaging Volume	0.175	m^3
Mounting Bracket	Galvanized + powder-coated, integrated	—
Solar Panel Orientation	South-facing	—
Adjustable Ground Angle	0 - 30	°
PV Cable Connector	Male/Female Aviation Connector	—
PV Cable Specification	2×1.5 National Standard, 80cm	—
DC Output Connector	5.5×2.1mm Male	—
DC Output Cable	2×0.75 Pure Copper, 1m	—

4.3 WakeCap Power 4800T (WP-4800T) Specifications

Model: [TBD] | 450W Solar, 4800Wh GEL Trailer Power Station

4.3.1 Solar Panel Specifications

Parameter	Value	Unit
Panel Type	Monocrystalline Silicon	—
Total Output Power	450	W
Panel Configuration	3 × 150W panels	—
[Additional specs]	[TBD]	—

4.3.2 Battery Specifications

Parameter	Value	Unit
Battery Type	GEL Deep Cycle (LEOCH brand)	—
Configuration	2 × 200Ah parallel	—
Total Capacity	400Ah / 4800Wh	—
System Voltage	12	V DC
Battery Brand	LEOCH	—

4.3.3 Controller Specifications

4.3.4 Mechanical Specifications

Parameter	Value	Unit
Total System Weight	525	kg
Packing Dimensions	2000 × 1400 × 2300	mm
Operating Dimensions	2000 × 2250 × 6500	mm
Mast Type	Pneumatic Telescoping	—
Mast Sections	5	—
Mast Material	Aluminum Alloy	—
Mast Height (Extended)	6.5	m
Mast Lifting Power	16W (negligible)	—
Spring Cable Specification	2×1.5	mm^2
Trailer Type	Single Axle	—
Trailer Standard	US Standard	—
Hitch Type	Ball Hitch	—
Tire Size	165/70R13	—
Outriggers	Manual	—
Surface Treatment	Anti-oxidation Electrostatic Powder Coat	—
Protection Level	IP65	—

4.3.5 Performance Specifications

Parameter	Value	Unit
Charging Time (0-100%)	~10	hours
Runtime (180W camera load)	~25	hours
Warranty	2	years
Camera Box Dimensions	With CCTV mounting box	—

4.4 WakeCap Power 21600T (WP-21600T) Specifications

Model: VTS3P | 1305W Solar, 21600Wh Lead Acid Trailer Power Station

4.4.1 Solar Panel Specifications

Parameter	Value	Unit
Number of Panels	3	—
Individual Panel Wattage	435	W
Total Output Power	1305	W
Panel Type	Monocrystalline Silicon	—

4.4.2 Battery Specifications

Parameter	Value	Unit
Battery Type	High Temperature Lead Acid	—
Configuration	6 × 12V 150Ah (3S2P for 24V)	—
Total Capacity	900Ah @ 24V	—
Energy Capacity	21,600	Wh
System Voltage	24	V DC

4.4.3 Controller Specifications

Parameter	Value	Unit
Controller Brand	EPEVER	—
Controller Type	MPPT	—
Current Rating	60	A

4.4.4 Mechanical Specifications

Parameter	Value	Unit
Total System Weight	1300	kg
Overall Dimensions	2320 × 1495 × 2350	mm
Mast Type	Manual Telescoping	—
Mast Height (Extended)	9	m
Mast Cables	2 × CAT6 + 2 × 1.5mm²	—
Camera Box Dimensions	412 × 412 × 416	mm
Trailer Type	Single Axle, Mechanical Brake	—
Suspension	Steel Plate Spring	—
Tire Size	15"	—
Tow Hitch	50mm Ball Hitch	—
Surface Treatment	Powder Coat (corrosion prevention)	—
Protection Level	IP65	—
Wind Rating	100	km/h
Container Loading	8 units per 40ft container	—

5. Solar Photovoltaic Subsystem

5.1 Solar Cell Technology

All WakeCap Power Solutions use monocrystalline silicon photovoltaic technology:

Product	Cell Grade	Efficiency	Technology
WP-384	Grade-A Monocrystalline	≥23%	Standard
WP-768	N-Type A+ Grade Monocrystalline	>22%	N-Type (better low-light)
WP-4800T	Monocrystalline	>20%	Standard
WP-21600T	Monocrystalline	>21%	Standard

5.2 Temperature Derating

Solar panel output decreases at elevated temperatures. Standard Temperature Coefficient (Pmax) is approximately -0.35%/°C above 25°C STC.

Example calculation for 50°C ambient:

- Temperature rise above STC: 50°C - 25°C = 25°C
- Cell temperature (add ~20°C for module): ~70°C
- Effective rise: 70°C - 25°C = 45°C
- Power reduction: 45 × 0.35% = 15.75%
- Effective panel output at 50°C: ~84% of rated

5.3 Panel Soiling Considerations

In desert environments, dust accumulation significantly impacts performance:

- Weekly cleaning recommended for optimal performance
- Dust buildup can reduce output by 15-25% within 2 weeks
- Tilt angle of 20-30° promotes natural dust shedding
- Use soft cloth and water only - no abrasive cleaners

5.4 Optimal Orientation

For maximum annual energy harvest in Saudi Arabia (latitude ~24°N):

- Azimuth: True South (0°)
- Tilt: 20-30° from horizontal

6. Battery Energy Storage Subsystem

6.1 Battery Chemistry Comparison

Parameter	$LiFePO_4$ (WP-384, WP-768)	GEL Lead Acid (WP-4800T)	Lead Acid (WP-21600T)
Nominal Voltage/Cell	3.2V	2.0V	2.0V
Cycle Life (80% DoD)	2000-5000 cycles	500-800 cycles	300-500 cycles
Calendar Life	8-10 years	5-7 years	3-5 years
Usable DoD	80-90%	50-70%	50-70%
Self-Discharge	<3%/month	3-5%/month	5-15%/month
Temperature Sensitivity	Low	Medium	High
Maintenance	None	Minimal	Periodic equalization
Weight (per Wh)	~8 Wh/kg	~35 Wh/kg	~35 Wh/kg
Cost (per Wh)	Higher	Medium	Lower
Safety	Excellent	Good	Good

6.2 $LiFePO_4$ Battery Management (WP-384, WP-768)

The integrated BMS provides comprehensive protection:

6.2.1 Protection Functions

- Over-charge protection: >14.6V (cuts charging)
- Over-discharge protection: <10V (disconnects load)
- Over-current protection: Configurable threshold
- Short-circuit protection: Instantaneous disconnect
- High-temperature protection: >60°C (inhibits charge/discharge)
- Low-temperature charge protection: <-10°C (inhibits charging)
- Cell balancing: Active balancing during charge

6.2.2 Cold Weather Operation

$LiFePO_4$ batteries have specific low-temperature behaviors:

Condition	Charging	Discharging
Above 0°C	Full rate	Full rate
-10°C to 0°C	Inhibited until recovery	Reduced capacity (~80%)
-20°C to -10°C	Inhibited	Reduced capacity (~60%)
Below -20°C	Inhibited	Inhibited

6.3 Lead Acid Battery Management (WP-4800T, WP-21600T)

6.3.1 Charging Profile

Lead acid batteries require a specific charging profile:

- Bulk Stage: Constant current at max rate until ~80% SOC
- Absorption Stage: Constant voltage (14.4V/cell) until current tapers
- Float Stage: Reduced voltage (13.5V/cell) for maintenance
- Equalization: Periodic overcharge (15.5V) to balance cells

6.3.2 Temperature Compensation

Charging voltage must be adjusted for temperature:

- Temperature coefficient: -3mV/°C per cell
- At 40°C: Reduce charge voltage by ~0.15V per cell
- At 50°C: Reduce charge voltage by ~0.25V per cell
- GEL batteries are more sensitive - follow manufacturer specs

6.4 Battery Maintenance Schedule

Task	$LiFePO_4$ (WP-384/60)	GEL (WP-4800T)	Lead Acid (WP-21600T)
Visual Inspection	Every 6 months	Monthly	Monthly
Terminal Cleaning	Annually	Quarterly	Quarterly
Voltage Check	Every 6 months	Monthly	Monthly
Capacity Test	Annually	Annually	Every 6 months
Equalization Charge	N/A	N/A (GEL)	Quarterly
Electrolyte Check	N/A (sealed)	N/A (sealed)	N/A (sealed)
Expected Replacement	8-10 years	5-7 years	3-5 years

7. Power Management and Control

7.1 MPPT Charge Controller

All systems use Maximum Power Point Tracking (MPPT) technology for optimal energy harvest.

7.1.1 MPPT Operating Principle

MPPT controllers continuously adjust the operating point of the solar array to extract maximum power:

- Monitors solar panel voltage and current
- Calculates instantaneous power ($P = V \times I$)
- Perturbs operating point to find maximum
- Converts voltage down to battery charging voltage
- Efficiency typically >95% (vs ~75% for PWM controllers)

7.2 Controller Specifications by Product

Parameter	WP-384	WP-768	WP-4800T	WP-21600T
Controller Type	Integrated MPPT	Standalone MPPT	MPPT	EPEVER MPPT
Rated Current	[TBD]	10A	30A	60A
Efficiency	>95%	96.5%	95%	98%
Max PV Input Voltage	~24V	20V	[TBD]	150V
Battery Voltage	12V	12V	12V	24V
Self-consumption	[TBD]	0.06W	[TBD]	[TBD]
Reverse Polarity Protection	Yes	Yes	Yes	Yes
Temperature Compensation	Yes	Yes	Yes	Yes
Communication	IIC/RS485	—	—	[TBD]

7.3 Output Regulation

DC output characteristics:

Parameter	WP-384	WP-768	WP-4800T	WP-21600T
Nominal Output Voltage	12V DC	10.5-12.6V DC	12V DC	24V DC
Voltage Tolerance	±0.5V	—	[TBD]	[TBD]
Max Continuous Current	3A	10A	<30A (25A typical)	<60A (55A typical)
Output Connector	Cable	5.5×2.1mm DC	[TBD]	[TBD]
Number of Channels	1	1	Multiple	Multiple

8. Protection and Safety Systems

8.1 Electrical Protection Features

WakeCap Power Solutions incorporate multiple layers of protection:

Protection Type	WP-384	WP-768	WP-4800T	WP-21600T
Over-Current Protection	4A (>3s)	10A	[TBD]	[TBD]
Short-Circuit Protection	>5A instant	Yes	Yes	Yes
Over-Voltage Protection	>14.6V	Yes	Yes	Yes
Under-Voltage Protection	<10V	Yes	Yes	Yes
Reverse Polarity Protection	Yes	Yes	Yes	Yes
Over-Temperature Protection	>60°C	Yes	[TBD]	[TBD]
Lightning/Surge Protection	[TBD]	[TBD]	[TBD]	[TBD]

8.2 Safety Warnings

⚠ DANGER

ELECTRICAL SHOCK HAZARD: Solar panels produce DC voltage when exposed to light. Disconnect all panels before servicing. Cover panels with opaque material if disconnection is not possible.

⚠ WARNING

BATTERY HAZARD: Lead-acid batteries (WP-4800T, WP-21600T) contain sulfuric acid and produce explosive hydrogen gas during charging. Ensure adequate ventilation. No smoking or open flames near batteries.

⚠ WARNING

LITHIUM BATTERY: $LiFePO_4$ batteries (WP-384, WP-768) must not be punctured, crushed, or exposed to fire. In case of damage, handle with care and dispose according to local regulations.

⚠ CAUTION

HOT SURFACES: Enclosures may reach temperatures exceeding 60°C in direct sunlight. Allow cooling before maintenance. Use appropriate PPE.

9. Environmental Performance

9.1 Ingress Protection

Product	IP Rating	Dust Protection	Water Protection
WP-384	IP66	Dust-tight	Powerful water jets
WP-768	IP66	Dust-tight	Powerful water jets
WP-4800T	IP65	Dust-tight	Low pressure water jets
WP-21600T	IP65	Dust-tight	Low pressure water jets

9.2 Temperature Specifications

Parameter	WP-384	WP-768	WP-4800T	WP-21600T
Operating Range	-20°C to +50°C	-30°C to +85°C	-20°C to +80°C	-20°C to +85°C
Storage Range	-40°C to +60°C	-30°C to +85°C	-20°C to +80°C	-20°C to +85°C
Optimal Performance	+10°C to +35°C	+10°C to +35°C	+10°C to +35°C	+10°C to +35°C

9.3 Corrosion Resistance

All systems are designed for harsh environments:

- Solar panel frames: Anodized aluminum alloy
- Mounting brackets: Galvanized + powder-coated steel
- Enclosures: UV-resistant polymers or coated metals
- Trailers: Anti-oxidation electrostatic powder coating
- Fasteners: Stainless steel or zinc-plated

For coastal deployments with salt air exposure:

- Increase inspection frequency to monthly
- Apply additional corrosion protection to exposed metal
- Consider supplementary enclosure sealing

10. Installation Engineering

10.1 Site Requirements

Requirement	WP-384	WP-768	WP-4800T	WP-21600T
Ground Area	N/A (pole mount)	N/A (structure mount)	~4 m²	~6 m²
Ground Type	Existing structure	Existing structure	Level, compacted	Level, compacted
Vehicle Access	Light vehicle	Light vehicle	Pickup with trailer	Heavy vehicle
Crane Required	No	No	No	No
Solar Exposure	Southern sky	Southern sky	Southern sky	Southern sky

10.2 Structural Requirements

10.2.1 WP-384 / WP-768 Mounting

- Pole diameter: 60-150mm (WP-384 clamps fit Ø150mm)
- Pole material: Steel or aluminum
- Minimum pole height: 3m above ground level
- Structural capacity: Must support system weight + wind load
- Wind load calculation: Consider local design wind speed

10.2.2 WP-4800T / WP-21600T Trailer Setup

- Level ground within 5° slope
- Deploy all outriggers (WP-4800T)
- Chock wheels before unhitching
- Anchor if wind speeds may exceed rating
- Minimum 2m clearance around trailer for maintenance

10.3 Electrical Installation

⚠ WARNING

All electrical connections must be made with system powered OFF. Verify zero voltage before connecting or disconnecting any cables.

10.3.1 Connection Sequence

Follow this sequence to prevent damage:

- 1. Connect load equipment (powered OFF)
- 2. Connect battery (WP-4800T/900T: connect batteries in correct series/parallel configuration)
- 3. Connect solar panels LAST

10.3.2 Disconnection Sequence

Reverse sequence for disconnection:

- 1. Cover solar panels with opaque material
- 2. Disconnect solar panels
- 3. Disconnect load equipment
- 4. Disconnect battery (if required for service)

11. Commissioning and Validation

11.1 Pre-Commissioning Checklist

□ All components received and undamaged

□ Installation completed per Section 10

□ Solar panels oriented correctly (south-facing, 20-30° tilt)

□ All electrical connections secure and properly terminated

□ No visible damage to cables, connectors, or enclosures

□ Battery state of charge verified (>40%)

11.2 System Startup

11.2.1 Initial Power-Up

- 1. Verify battery voltage within normal range
- 2. Connect solar panels (during daylight)
- 3. Verify charging indicator active (blue LED flashing on WP-384)
- 4. Measure output voltage at load terminals
- 5. Connect load equipment
- 6. Verify load operates correctly

11.3 Validation Tests

Test	Expected Result	Pass Criteria
Battery Voltage	Within rated range	WP-384/60: 10.5-14.8V, WP-21600T: 22-29V
Charging Current	Positive during sunlight	>0.5A with sun
Output Voltage	Nominal ± tolerance	WP-384: 12V ±0.5V
Load Operation	Equipment powers ON	All devices functional
Indicator Status	Normal operation	No fault indicators
Night Operation	Battery supplies load	No interruption

11.4 Documentation

Record the following for system handover:

- Serial numbers of all major components

- Installation date and location coordinates
- Initial battery voltage and state of charge
- Connected load equipment and power consumption
- Photos of installation (overview, connections, solar orientation)

12. Maintenance Engineering

12.1 Maintenance Schedule

Task	Frequency	WP-384/60	WP-4800T/900T
Visual inspection	Monthly	✓	✓
Solar panel cleaning	Weekly-Monthly	✓	✓
Connection tightness check	Quarterly	✓	✓
Battery voltage check	Monthly	—	✓
Enclosure seal inspection	Quarterly	✓	✓
Trailer wheel/brake check	Monthly	N/A	✓
Mast mechanism inspection	Monthly	N/A	✓
Full system test	Annually	✓	✓

12.2 Solar Panel Maintenance

- Clean with soft cloth and water only
- No abrasive materials or solvents
- Inspect for cracks, delamination, or discoloration
- Check cable integrity and connector condition
- Verify mounting bracket tightness

12.3 Battery Maintenance

12.3.1 $LiFePO_4$ Batteries (WP-384, WP-768)

- No routine maintenance required
- Keep terminals clean and dry
- Monitor SOC indicator for anomalies
- Store at 40-60% SOC if not in use for extended periods

12.3.2 Lead Acid Batteries (WP-4800T, WP-21600T)

- Check terminal corrosion monthly
- Clean terminals with baking soda solution if corroded
- Verify electrolyte level (if accessible)
- Perform equalization charge quarterly (standard lead acid only)
- Replace batteries showing capacity below 80%

13. Troubleshooting and Diagnostics

13.1 Symptom-Based Troubleshooting

Symptom	Possible Cause	Diagnostic Steps	Solution
No output power	Battery depleted	Check battery voltage	Recharge in sunlight
No output power	Over-current trip	Check load current	Reduce load, reset system
No charging	Panel disconnected	Check panel connections	Reconnect panels
No charging	Panel shaded	Inspect panel surface	Remove shade source
No charging	Controller fault	Check controller LED	Replace controller
Low output voltage	Battery worn	Capacity test	Replace battery
Intermittent power	Loose connection	Inspect all terminals	Tighten connections
Overheating	Overload	Measure load current	Reduce load
Overheating	Poor ventilation	Check enclosure temp	Improve airflow

13.2 LED Diagnostic Reference (WP-384)

LED Pattern	Meaning	Action
Blue flashing	Charging active	Normal operation
Blue steady	Charge complete	Normal operation
Blue off	No solar input	Check panels/sun
Green steady	Output normal	Normal operation
Green flashing	Overload/short	Reduce load, check wiring
Green off	Shutdown/fault	Check battery, reset
Red flashing	Low battery	Recharge system
Red off	Battery OK	Normal operation

13.3 When to Escalate

Contact technical support if:

- Battery fails to hold charge after full recharge cycle
- Controller shows persistent fault even after reset
- Physical damage to enclosures, panels, or structure
- System repeatedly trips protection without apparent cause
- Performance degrades significantly (>30% below expected)

14. Appendices

Appendix A: Glossary

Term	Definition
Ah (Ampere-hour)	Unit of battery capacity; 1Ah = 1 ampere for 1 hour
BMS	Battery Management System - monitors and protects battery pack
DoD	Depth of Discharge - percentage of capacity used from battery
$LiFePO_4$	Lithium Iron Phosphate - a lithium battery chemistry
MPPT	Maximum Power Point Tracking - optimizes solar energy harvest
PSH	Peak Sun Hours - equivalent hours of 1000 W/m^2 irradiance
SOC	State of Charge - remaining battery capacity percentage
Vmp	Voltage at Maximum Power point
Voc	Open Circuit Voltage - panel voltage with no load
Wh (Watt-hour)	Unit of energy; 1Wh = 1 watt for 1 hour

Appendix B: Contact Information

Technical Support: [TBD]

Spare Parts: [TBD]

Emergency: [TBD]

Appendix C: Document References

- WC-PS-LRG-v1.0: Power Solutions Logistics Reference Guide
- WC-PS-OSG-v1.0: Power Solutions Operations Selection Guide
- [TBD]: Installation Drawings
- [TBD]: Wiring Diagrams

END OF DOCUMENT

WakeCap Power Solutions - Technical Engineering Manual v1.2